A COMPANION NOTE KIRMSE–COXETER AND THE σ-TWIST, BY CONCRETE EXAMPLE

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Abstract. This companion note was written by the AI agent (drafted by Claude Opus 4.6 in max-thinking mode, revised 23 May 2026 by Claude Opus 4.7, Anthropic) at the direction of Jens Köplinger, the author of the accompanying paper "An order on the Leech lattice from transposition-twisted octonions." It is a *training document*, assembled from the overall corpus of the project repository `bitbucket.org/jenskoeplinger/leech_alg`, intended as a learning tool for the principal investigator and for other readers who approach this construction with the kind of questions – and, candidly, misconceptions – that arose throughout the investigation. Authorship and intellectual leadership of the main paper and the research programme belong to Köplinger, with AI collaboration as acknowledged there; the present companion is solely the AI agent's rendering of that work in expository form and makes no claim to new mathematics.

The note walks through the construction step by step, pairing every abstract definition with a concrete hand-checkable example, first at the E_8 level (where Coxeter's correction of Kirmse provides the historical precedent) and then at the Leech level (where the σ-twist plays the analogous role). Readers who are comfortable with the main paper will find nothing new here; readers who want to see every piece verified by hand will find the explicit numerics they may have missed.

1. Preliminaries

This note follows the conventions of the main paper. We work with the real octonion algebra $\mathbb{O}$ on $\mathbb{R}^8$ with basis $\{e_0, e_1, \ldots, e_7\}$, and with Wilson's construction of the Leech lattice inside the triple octonion space $\mathbb{R}^{24} = \mathbb{O}_1 \oplus \mathbb{O}_2 \oplus \mathbb{O}_3$.

The guiding principle throughout is: *distinguish abstract structure from the specific lattice to which it is applied.* The same ambient algebra can host many lattices; the same lattice can be tested against many bilinear products on the ambient space. "Closure of this lattice under this product" is a question about both, not just one.

2. The E_8 lattice

Abstract. The lattice E_8 is the unique even unimodular lattice in $\mathbb{R}^8$ (up to isomorphism). "Even" means every vector $v \in E_8$ has integer squared norm

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$\|v\|^2 \in 2\mathbb{Z}$. "Unimodular" means E_8 equals its own dual lattice (the set of vectors having integer inner product with every element of E_8). It contains 240 vectors of minimal squared norm 2, the *roots*.

Concrete. Following Wilson, take for L the integer span of the following 240 octonions:

- **112 "type-1" roots** of the form $\pm e_a \pm e_b$ with $0 \le a < b \le 7$;
- **128 "type-2" roots** of the form $\frac{1}{2}(\varepsilon_0 e_0 + \varepsilon_1 e_1 + \cdots + \varepsilon_7 e_7)$ where each $\varepsilon_i \in \{+1, -1\}$ and the number of -1s is odd.

Equivalently,

$$L = \{v \in \mathbb{Z}^8 : \sum_i v_i \text{ is even}\} \;\cup\; \{v \in (\mathbb{Z} + \tfrac{1}{2})^8 : \sum_i v_i \text{ is odd}\}.$$

Example 2.1. *The vector* $(1,1,0,0,0,0,0,0) = e_0 + e_1$ *is a type-1 root: integer coordinates, sum* 2. *The vector* $\frac{1}{2}(-1,1,1,1,1,1,1,1)$ *is a type-2 root: half-integer coordinates, one* -1 *(odd count), coordinate sum* 3 *(odd). Both lie in* L*; both have squared norm* 2.

Example 2.2. *Distinguished element in* L *to be used throughout:*

$$s \;=\; \tfrac{1}{2}(-e_0 + e_1 + e_2 + e_3 + e_4 + e_5 + e_6 + e_7),$$

of squared norm 2. *In coordinates* $s = (-\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})$*: half-integer, one* $-1/2$ *(odd count), sum* 3 *(odd). Confirmed in* L.

3. The octonion algebra

Abstract. The octonions $\mathbb{O}$ are the unique real *normed division algebra* of dimension 8: a real 8-dimensional vector space with a bilinear product $x \cdot y$ and a positive-definite quadratic norm satisfying $\|x \cdot y\| = \|x\| \cdot \|y\|$. The algebra is non-commutative and non-associative (but alternative).

Concrete. Fix the basis $\{e_0, e_1, \ldots, e_7\}$ with e_0 the multiplicative identity. The multiplication on the imaginary units is determined by the seven *Fano triples*

$$(1,2,4),\ (2,3,5),\ (3,4,6),\ (4,5,7),\ (5,6,1),\ (6,7,2),\ (7,1,3), \tag{1}$$

read cyclically: for (a,b,c) above, $e_a \cdot e_b = +e_c$, $e_b \cdot e_c = +e_a$, $e_c \cdot e_a = +e_b$, with opposite signs for the reversed products. Every imaginary unit squares to $-e_0$.

Example 3.1. *From the triple* $(1,2,4)$*:* $e_1 \cdot e_2 = +e_4$ *and* $e_2 \cdot e_1 = -e_4$. *From* $(7,1,3)$*:* $e_1 \cdot e_3 = +e_7$. *From* $(6,7,2)$*:* $e_7 \cdot e_2 = +e_6$*, so* $e_2 \cdot e_7 = -e_6$.

Closure of L under the octonion product. Under the Fano convention (1), the lattice L is closed under octonion multiplication: for every $u, v \in L$, the product $u \cdot v \in L$. This is a classical result of Coxeter (1946), verified in the main paper by computing all $8 \times 8 = 64$ products of a $\mathbb{Z}$-basis of L in exact arithmetic and checking that each lies again in L.

Example 3.2. *A sample product. From $s \cdot s$ with s as above, using the identity $(\sum_{k=1}^{7} e_k)^2 = -7e_0$:*

$$s \cdot s \;=\; -\tfrac{3}{2}\, e_0 - \tfrac{1}{2}(e_1 + e_2 + e_3 + e_4 + e_5 + e_6 + e_7).$$

Its coordinates are all half-integer; the coordinate sum is $-\frac{3}{2} - 7 \cdot \frac{1}{2} = -5$, odd. Hence $s \cdot s \in L$.

4. The Kirmse–Coxeter precedent

What Kirmse did, and where it went wrong. Kirmse (1924) fixed one octonion multiplication – call it $\cdot_K$ – given by the seven Fano triples

$$(1,2,3),\ (1,5,4),\ (1,6,7),\ (2,4,7),\ (2,6,5),\ (3,4,6),\ (3,5,7)$$

(read cyclically, as in (1); this labelling differs from (1)). He then looked for the *maximal orders*: the largest lattices in $(\mathbb{R}^8, \cdot_K)$ closed under multiplication. His table $\cdot_K$ is itself a perfectly correct octonion algebra – that is *not* where the error lies. He listed eight candidate maximal orders and described one of them, J_1, in detail.

J_1 is the lattice spanned over $\mathbb{Z}$ by the eight units $1, i_1, \ldots, i_7$ together with the four half-integer octonions

$$a_1 = \tfrac{1}{2}(1{+}i_1{+}i_2{+}i_3), \qquad a_2 = \tfrac{1}{2}(i_4{+}i_5{+}i_6{+}i_7),$$
$$a_3 = \tfrac{1}{2}(1{+}i_1{+}i_4{+}i_5), \qquad a_4 = \tfrac{1}{2}(1{+}i_2{+}i_4{+}i_7).$$

Kirmse's error is that J_1 is *not closed* under $\cdot_K$. The witness, found by Coxeter (1946) and reproduced in exact arithmetic in `verify_coxeter_1946.py`, is

$$a_1 \cdot_K a_3 = \tfrac{1}{2}(1{+}i_1{+}i_2{+}i_3) \cdot_K \tfrac{1}{2}(1{+}i_1{+}i_4{+}i_5) = \tfrac{1}{2}(i_1{+}i_2{+}i_4{+}i_7),$$

and $\frac{1}{2}(i_1{+}i_2{+}i_4{+}i_7) \notin J_1$. So J_1 is not an order at all. There are in fact *seven* maximal orders for $\cdot_K$, not eight; J_1 is the spurious entry on Kirmse's list. Coxeter identified the mistake; R. H. Bruck supplied the remedy.

The remedy: a transposition of two coordinate axes. Bruck's correction, as Coxeter reports it, is to "transpose two of the i's." Concretely: pick a transposition of two imaginary basis axes – a linear involution of $\mathbb{R}^8$ that swaps $e_a \leftrightarrow e_b$ and fixes the other six. *Any* of the $\binom{7}{2} = 21$ such transpositions repairs J_1 (Coxeter; the 21 fall into 7 triples of equal effect, giving the 7 maximal orders – verified).

We write σ for such a transposition, the *same* symbol the main paper uses – because it is the same kind of map, and disguising that with two symbols would only obscure the point of this note. What differs between the Kirmse–Coxeter use and the main paper's use is never the map: it is the *algebra and lattice the map is applied to*, and we keep that explicit at every step. For definiteness take $\sigma = (1\ 2)$, the swap $e_1 \leftrightarrow e_2$ – which is at once one of the 21 transpositions that repair Kirmse's J_1 and the transposition the main paper twists by.

The repair has two equivalent descriptions, and it is worth being completely explicit about *which object each one moves.*

Framing A – σ acts on the lattice (on the vector space). Keep the product $\cdot_K$. Replace the lattice J_1 by its image $J := \sigma(J_1)$. Then $(J, \cdot_K)$ *is* a maximal order: J is closed under $\cdot_K$. The algebra is untouched; what changed is the subset of $\mathbb{R}^8$ being tested.

Framing B – σ acts on the product (on the algebra). Keep the lattice J_1. Replace $\cdot_K$ by its σ-conjugate

$$a \cdot_K^\sigma b \;:=\; \sigma\big(\sigma(a) \cdot_K \sigma(b)\big).$$

Then $(J_1, \cdot_K^\sigma)$ is a maximal order: J_1 is closed under $\cdot_K^\sigma$. The lattice is untouched; what changed is the multiplication.

These are not two corrections but one. σ is a single linear map, and $\sigma\colon (\mathbb{R}^8, \cdot_K^\sigma) \to (\mathbb{R}^8, \cdot_K)$ is an algebra *isomorphism*, so it carries closed sublattices to closed sublattices: J_1 is closed under $\cdot_K^\sigma$ *iff* $\sigma(J_1)$ is closed under $\cdot_K$. Because σ is orthogonal, $\cdot_K^\sigma$ is again a *composition* algebra – again an octonion algebra; the "corrected table" is just another octonion table, isomorphic to Kirmse's via σ.

Example 4.1 (What the Kirmse–Coxeter twist modifies)**.** *Take $\sigma = (1\ 2)$. In framing B it rewrites exactly those basis products that involve a swapped axis, and leaves the rest alone (verified in `verify_coxeter_1946.py`):*

$$\begin{aligned}
i_1 \cdot_K i_2 &= +i_3 &\quad&\rightsquigarrow&\quad i_1 \cdot_K^\sigma i_2 &= -i_3,\\
i_1 \cdot_K i_5 &= +i_4 &&\rightsquigarrow& i_1 \cdot_K^\sigma i_5 &= -i_6,\\
i_6 \cdot_K i_7 &= +i_1 &&\rightsquigarrow& i_6 \cdot_K^\sigma i_7 &= +i_2,\\
i_3 \cdot_K i_4 &= +i_6 &&\rightsquigarrow& i_3 \cdot_K^\sigma i_4 &= +i_6 \quad (\textit{unchanged}).
\end{aligned}$$

And on the very pair that broke closure – the same two elements a_1, a_3 of the same lattice J_1:

$$a_1 \cdot_K a_3 = \tfrac{1}{2}(i_1+i_2+i_4+i_7) \notin J_1, \qquad a_1 \cdot_K^\sigma a_3 = \tfrac{1}{2}(i_1+i_3+i_4+i_7) \in J_1.$$

Twisting the product (framing B) restored closure without moving the lattice.

Observation. *A convention of octonion multiplication fixes how basis products are assigned. A different convention, obtained by transposing two basis indices, assigns products differently. For some lattices, the two conventions are interchangeable (the lattice is closed under both); for others, only one convention closes. Which lattices behave which way is precisely the question that Kirmse got wrong and Coxeter got right.*

A concrete lattice where the story does matter. From here on we return to the multiplication $\cdot$ of (1) – one of Coxeter's seven corrected octonion tables – and to the lattice L, which closes under it.

Whether a transposition twist changes anything depends entirely on the lattice it is asked about. For the lattice L itself the twist is invisible: L is symmetric under every coordinate permutation, so transposing two axes produces an algebra under which L also closes (this is made precise in Section 10). Kirmse's J_1, by contrast, is *not* symmetric under such a swap – which is exactly why the twist moved it. A second lattice of that non-symmetric kind, the one that drives the rest of this note, appears naturally in Wilson's Leech construction: the sublattice

$$Ls \;=\; \{\ell \cdot s : \ell \in L\},$$

the image of L under right-multiplication by the element s above. This is again a full-rank sublattice of L, of index 16 (see the main paper, Section 2.3). It is a lattice, so of course it is closed under addition; the question is whether it is closed under octonion multiplication.

Example 4.2 (Standard product fails on Ls)**.** *Take*

$$a = (-1,\, 0,\, 0,\, 0,\, 1,\, 0,\, 1,\, 1), \qquad b = (-1,\, 1,\, 0,\, 0,\, 0,\, 1,\, 0,\, 1).$$

Both a and b are in Ls (they appear as basis vectors of Ls produced in the main paper's exact-arithmetic verification). Their octonion product, computed under the Fano convention (1)*, is*

$$a \cdot b = (0,\, -2,\, 2,\, 1,\, -2,\, -1,\, -1,\, -1).$$

This vector has integer coordinates and lies in L – but it is not *an integer combination of a $\mathbb{Z}$-basis of Ls. The verification is carried out with exact rational arithmetic in the script* `symbolic_proof_checks.py`*.*

Consequently, $Ls \cdot Ls \not\subseteq Ls$.

This is the structural echo of Kirmse's situation at the sublattice level: a specific sublattice, under a specific octonion product, fails to close.

5. The σ-twist, abstractly

Abstract. Fix a transposition $\sigma = (s\ t)$ on the imaginary basis indices $\{1, \ldots, 7\}$ (for example, $\sigma = (1\ 2)$). Extend σ linearly to a map $\mathbb{R}^8 \to \mathbb{R}^8$ that fixes e_0 and swaps $e_s \leftrightarrow e_t$. This σ is an *isometry* of $\mathbb{R}^8$: it preserves the Euclidean inner product. It is also an involution: $\sigma^2 = \mathrm{id}$.

Starting from the standard product $\cdot$, the map σ induces a *σ-twisted product* on the same $\mathbb{R}^8$:

$$x \cdot_\sigma y \;=\; \sigma\big(\sigma(x) \cdot \sigma(y)\big). \tag{2}$$

The pair $(\mathbb{R}^8, \cdot_\sigma)$ is again an octonion algebra, abstractly isomorphic to $(\mathbb{R}^8, \cdot)$ via σ itself. But it is a *different bilinear product* on the same $\mathbb{R}^8$: it assigns different specific products to pairs of basis vectors.

Concrete. Take $\sigma = (1\ 2)$, which swaps e_1 and e_2 (and fixes $e_0, e_3, e_4, e_5, e_6, e_7$). Then

$$e_1 \cdot_\sigma e_2 = \sigma(\sigma(e_1) \cdot \sigma(e_2)) = \sigma(e_2 \cdot e_1) = \sigma(-e_4) = -e_4,$$

whereas $e_1 \cdot e_2 = +e_4$ under the standard product. Similarly one checks $e_1 \cdot_\sigma e_4 = +e_2$ (vs. $-e_2$ under $\cdot$), and so on: the two products disagree on some pairs of basis elements.

What is twisted and what is invariant.

- **Twisted:** the multiplication table. Under $\cdot_\sigma$, the Fano triples (1) are replaced by $(\sigma(a), \sigma(b), \sigma(c))$ for each original triple.
- **Invariant under σ (as a linear map):** the underlying vector space $\mathbb{R}^8$, the Euclidean inner product, the decomposition into integer and half-integer vectors, and the lattice L itself ($\sigma(L) = L$ since σ is a coordinate permutation).
- **Not invariant under σ:** specific sublattices of L that depend on a distinguished element, such as Ls. Since Ls is defined as $\{\ell \cdot s : \ell \in L\}$, applying σ gives
$$\sigma(Ls) = \{\sigma(\ell) \cdot_\sigma \sigma(s) : \ell \in L\} \;\neq\; Ls$$
in general. The main paper verifies $\sigma(Ls) \neq Ls$ directly: for $\sigma = (1\ 2)$, the vector $(-1, 0, 1, 0, 0, 1, 0, 1)$ lies in $\sigma(Ls)$ but not in Ls.

The last two bullets together say: *σ is an isometry of $\mathbb{R}^8$ that fixes L but moves Ls to a distinct sublattice.*

6. The twist at the E_8 level changes the story

The sublattice $\sigma(Ls)$ is the image of Ls under the transposition. It has the same rank as Ls, the same index in L, and is isometric to Ls. But its elements are different vectors. The question of closure under the standard octonion product is now posed afresh.

Example 6.1 (Standard product succeeds on $\sigma(Ls)$)**.** *A $\mathbb{Z}$-basis of $\sigma(Ls)$ is obtained by applying σ (coordinate swap of positions $1, 2$) to each basis vector of Ls. Using the basis generated in the main paper's verification, all $8 \times 8 = 64$ products of basis vectors of $\sigma(Ls)$, under the standard octonion product $\cdot$, lie again in $\sigma(Ls)$. The verification is in `symbolic_proof_checks.py` as Lemma D.*

Hence $\sigma(Ls) \cdot \sigma(Ls) \subseteq \sigma(Ls)$.

Two equivalent readings. Because σ is an isometry and an algebra isomorphism $(\mathbb{R}^8, \cdot) \to (\mathbb{R}^8, \cdot_\sigma)$, one can restate Example 6.1 in an equivalent form on the original lattice Ls with the twisted product:

$$Ls \cdot_\sigma Ls \;\subseteq\; Ls.$$

(Apply σ to both sides of the inclusion $\sigma(Ls) \cdot \sigma(Ls) \subseteq \sigma(Ls)$ and use $\sigma^2 = \mathrm{id}$ together with (2).) The two forms are tautologically equivalent.

Reading both against Example 4.2:

$$\underbrace{Ls \cdot Ls \not\subseteq Ls}_{\text{standard product fails on } Ls} \qquad \text{versus} \qquad \underbrace{Ls \cdot_\sigma Ls \subseteq Ls}_{\text{twisted product succeeds on } Ls} .$$

Equivalently:

$$\underbrace{Ls \cdot Ls \not\subseteq Ls}_{Ls \text{ fails}} \qquad \text{versus} \qquad \underbrace{\sigma(Ls) \cdot \sigma(Ls) \subseteq \sigma(Ls)}_{\sigma(Ls) \text{ succeeds}}.$$

Interpretation. Two different bilinear products on $\mathbb{R}^8$ (namely $\cdot$ and $\cdot_\sigma$) are isomorphic as abstract algebras. Every theorem phrased in the language of the abstract algebra carries over. But "does lattice X close under product $\star$?" is not a theorem phrased purely in the abstract algebra: it also names a specific subset $X \subset \mathbb{R}^8$, and the answer depends on how the isomorphism relates X to itself.

If $\sigma(X) = X$, the answer is the same for both products. If $\sigma(X) \neq X$, the answer can flip: X may fail to close under $\cdot$ while $\sigma(X)$ closes under $\cdot$; equivalently, X closes under $\cdot_\sigma$ while $\sigma(X)$ fails under $\cdot_\sigma$. The lattice L itself is σ-invariant and therefore closes under both products. The sublattice Ls is *not* σ-invariant, and this is exactly what opens the gap between the two closure statements.

This is the structural lesson of Kirmse–Coxeter in isolation. The same lesson, applied one dimension higher, is the content of the main paper.

7. Wilson's Leech lattice

Abstract. The Leech lattice Λ is the unique even unimodular lattice of rank 24 with no roots (i.e. no vectors of squared norm 2). Its minimal vectors have squared norm 4, and there are exactly 196,560 of them.

Concrete (Wilson's construction). Identify $\mathbb{R}^{24}$ with $\mathbb{O}_1 \oplus \mathbb{O}_2 \oplus \mathbb{O}_3$: three copies of the octonions, labelled $\alpha = 1, 2, 3$. Let $\bar{s} = \frac{1}{2}(-e_0 - e_1 - e_2 - \cdots - e_7)$, the octonion conjugate of the negative of the earlier s, again of squared norm 2. Note $\bar{s} \notin L$ under the lattice membership criterion of Section 2: its coordinates are half-integer but its coordinate sum is -4, which falls in the excluded even branch of the half-integer case. Define two sublattices of L:

$$Ls = \{\ell \cdot s : \ell \in L\}, \qquad L\bar{s} = \{\ell \cdot \bar{s} : \ell \in L\},$$

each of index 16 in L. These sets are sublattices of L despite $\bar{s} \notin L$: for every $\ell \in L$ the product $\ell \cdot \bar{s}$ nonetheless lies in L (verified directly on the 240 roots and extended by linearity). Wilson's Leech lattice is then

$$\Lambda = \big\{(x, y, z) \in L^3 : \; x + y,\ x + z,\ y + z \in L\bar{s}; \quad x + y + z \in Ls\big\}. \tag{3}$$

The sublattice Ls from Section 4 is precisely the "condition-(3) block sum" lattice in Wilson's definition.

8. The standard triple product on $\mathbb{R}^{24}$

Abstract. The natural bilinear product on $\mathbb{R}^{24} = \mathbb{O}_1 \oplus \mathbb{O}_2 \oplus \mathbb{O}_3$ built from a single octonion product is a *triple product with $\mathbb{Z}_3$-symmetric cross-block routing.* For $u = (x, y, z)$ and $v = (x', y', z')$ define

$$u \star v \;=\; (P_1, P_2, P_3),$$

where

$$\begin{aligned} P_1 &= x \cdot x' + z \cdot y' + y \cdot z', \\ P_2 &= y \cdot y' + x \cdot z' + z \cdot x', \\ P_3 &= z \cdot z' + y \cdot x' + x \cdot y'. \end{aligned}$$

Same-block products stay in the block; cross-block products $O_\alpha \times O_\beta$ land in O_γ with $\{\alpha, \beta, \gamma\} = \{1, 2, 3\}$. The three blocks play the same role under cyclic rotation $1 \to 2 \to 3 \to 1$.

Why condition (3) is the obstruction. A direct computation from the definitions gives

$$P_1 + P_2 + P_3 \;=\; (x + y + z) \cdot (x' + y' + z').$$

Thus "$u \star v$ satisfies Wilson's condition (3)" amounts to: "the product of two elements of Ls lies in Ls".

Using $u, v \in \Lambda$ means $x + y + z \in Ls$ and $x' + y' + z' \in Ls$, so the question reduces exactly to whether $Ls \cdot Ls \subseteq Ls$. By Example 4.2, it does not. Hence:

Example 8.1 (Standard triple product fails on Λ)**.** *There exist $u, v \in \Lambda$ for which $u \star v \notin \Lambda$: the block sum of $u \star v$ need not lie in Ls, failing Wilson's condition (3). The concrete obstruction is precisely the pair (a, b) of Example 4.2: any $u, v \in \Lambda$ whose triple sums equal a, b respectively produce a $u \star v$ whose block sum equals $a \cdot b \notin Ls$.*

9. The σ-twisted triple product

Abstract. Apply σ componentwise to each of the three octonion blocks. This yields a linear map on $\mathbb{R}^{24}$, still denoted σ, that is an isometry and satisfies $\sigma^2 = \mathrm{id}$. Replace the octonion product in each of the nine block-pair products of $\star$ by the σ-twisted product $\cdot_\sigma$ from (2): this produces a new bilinear product $\star_\sigma$ on $\mathbb{R}^{24}$. By the same formal manipulation as before,

$$u \star_\sigma v \;=\; \sigma\big(\sigma(u) \star \sigma(v)\big),$$

where on the right $\star$ is the standard (untwisted) triple product.

Concrete. Choose $\sigma = (1\ 2)$, acting on imaginary indices of each octonion block. Then for $u, v \in \Lambda$,

$$(\text{block sum of } u \star_\sigma v) \;=\; \sigma\big((\sigma(x) + \sigma(y) + \sigma(z)) \cdot (\sigma(x') + \sigma(y') + \sigma(z'))\big).$$

Inside the outer σ, each triple sum lies in $\sigma(Ls)$ (since each original triple sum lies in Ls, by Wilson's condition (3) on u, v). By Example 6.1, $\sigma(Ls)\cdot\sigma(Ls) \subseteq \sigma(Ls)$. Applying the outer σ brings the result into $\sigma(\sigma(Ls)) = Ls$.

So the block sum of $u \star_\sigma v$ lies in Ls: Wilson's condition (3) is restored. Conditions (1) and (2) go through by analogous arguments (see the main paper, Section 4). The net effect is:

Example 9.1 (σ-twisted triple product succeeds on Λ)**.** *For all $u, v \in \Lambda$, the product $u \star_\sigma v$ lies in Λ. This is Theorem 3.1 of the main paper.*

What is twisted and what is invariant (at the Leech level).

- **Twisted:** the octonion multiplication used inside each of the nine block-pair products. Equivalently, the Fano triples (1) are replaced block by block by their σ-images.
- **Invariant under σ:** the ambient space $\mathbb{R}^{24}$, the three-block decomposition into $\mathbb{O}_1, \mathbb{O}_2, \mathbb{O}_3$, the $\mathbb{Z}_3$ cross-block routing rule, and the sub-lattice L of each block ($\sigma(L) = L$).
- **Not invariant under σ:** the sub-lattice Ls of each block ($\sigma(Ls) \neq Ls$), and through it the Leech lattice itself ($\sigma(\Lambda) \neq \Lambda$ in general, since Λ is defined by a condition involving Ls).

The comparison with the E_8 level is exact. One level down, σ fixes L but not Ls, and this is what allows the twist to change the closure question for Ls. One level up, σ fixes $\mathbb{O}_1 \oplus \mathbb{O}_2 \oplus \mathbb{O}_3$ but not Λ, and this is what allows the twist to change the closure question for Λ.

10. Could the σ-twist undo Coxeter's fix?

A natural concern. Coxeter's correction of Kirmse and the present σ are both transpositions of two imaginary coordinate axes – the same kind of map, a linear involution of $\mathbb{R}^8$ (Section 4); indeed $\sigma = (1\ 2)$ is one of the 21 transpositions that repair Kirmse's J_1. Could a poor choice of σ, then, reverse Coxeter's correction and return the E_8 lattice to a non-closing regime?

It cannot, and the reason is one short computation.

Proposition. *For every coordinate transposition σ, the lattice $L = D_8^+$ remains closed under the twisted product: $L \cdot_\sigma L \subseteq L$.*

Proof. $L = D_8^+$ is invariant under every coordinate permutation, so $\sigma(L) = L$. The main paper establishes $L \cdot L \subseteq L$. Hence

$$L \cdot_\sigma L = \sigma\big(\sigma(L) \cdot \sigma(L)\big) = \sigma(L \cdot L) \subseteq \sigma(L) = L. \qquad \square$$

So no σ-twist can disturb the E_8-level closure. This is exactly Wilson's condition (1): for $u, v \in \Lambda \subseteq L^3$, every block of $u \star_\sigma v$ is a sum of twisted products $x \cdot_\sigma x'$ with octonion factors $x, x' \in L$, and the proposition returns each block to L. Condition (1) holding *is* Coxeter's closure carried through the twist; the construction depends on the correction, it does not undo it.

The twist does act non-trivially – but only on the lattices σ does *not* fix. L is fixed; the sublattices Ls and $L\bar{s}$ are not, and that is where Wilson's conditions (2) and (3) are redirected. There is no tension between "σ fixes L" and "σ moves Kirmse's J_1": L and J_1 are both copies of E_8, but *placed* differently in $\mathbb{R}^8$ – $L = D_8^+$ is the coordinate-symmetric placement, J_1 is not (it is fixed by none of the 21 transpositions). Whether a transposition fixes a lattice is a property of the lattice's placement, not of the map. Taking the order to be the symmetric placement D_8^+ is exactly what makes the twist free at the E_8 level; Coxeter's transposition, asked instead about the non-symmetric J_1, necessarily moves it.

11. Summary of concrete data

Three empirical facts are verified by exact rational arithmetic in `symbolic_proof_checks.py`:

(1) $\sigma(Ls) \neq Ls$ (explicit witness: $(-1, 0, 1, 0, 0, 1, 0, 1) \in \sigma(Ls) \setminus Ls$).
(2) $Ls \cdot Ls \not\subseteq Ls$ under the standard octonion product (explicit witness a, b in Example 4.2 with $a \cdot b \notin Ls$).
(3) $\sigma(Ls) \cdot \sigma(Ls) \subseteq \sigma(Ls)$ under the standard octonion product (all 64 basis products land in $\sigma(Ls)$).

Two equivalent phrasings of the main result are:

- "Λ is closed under the σ-twisted triple product $\star_\sigma$."
- "$\sigma(\Lambda)$ is closed under the standard triple product $\star$."

Both say the same thing via the algebra isomorphism $\sigma : (\mathbb{R}^{24}, \star) \to (\mathbb{R}^{24}, \star_\sigma)$; neither says that Λ is closed under $\star$, which is false.

Anthropic